PROPER BASE CHANGE FOR HIGHER DIRECT IMAGES OF A MODULE ON A RINGED SPACE, COMPUTED ON THE FIBRE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is complete as written and ends with the construction of the neighbourhood V; no step is referred to the reader.

Editorial note. The counted spans carry 5 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-cohomology/retained-preamble.tex lines 1-81), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-proper-base-change also occurs in etale-cohomology.tex, but there the statement concerns base change for Rf_* of a complex with torsion cohomology on the etale site of a proper scheme; hypotheses and conclusion are different, so only the label coincides and this card counts its own proposition.

Allowed standard prerequisites (editorial).

- Base change map for higher direct images. Same chapter, Lemma lemma-base-change-map-flat-case, invoked at proof lines 3073-3074. Supplies the canonical map whose bijectivity is proved.
- A criterion making the restriction of an injective to a closed subset acyclic. Same chapter, Lemma lemma-cohomology-of-closed, invoked at proof lines 3077-3081 and again at 3091-3092. Supplies that the restrictions of the injective resolution to the fibre are acyclic for global sections, and the identification of the resulting complex.
- Leray acyclicity: a bounded below complex of acyclic objects computes the derived functor. Derived Categories, Lemma derived-lemma-leray-acyclicity, invoked at proof lines 3082-3084. Supplies that the complex of sections over the fibre represents the derived global sections.

Source reference. Original lemma 015E: sources/stacks-chow/tags.txt, source lines 1505–1505. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02N7: sources/stacks-cohomology/pinned-original-source.tex, source lines 2977–2977. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 09V3: sources/stacks-cohomology/pinned-original-source.tex, source lines 2848–2848. Pointer outside the counted span, invoked once

Date: stacks-project:cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0CY2: sources/stacks-chow/tags.txt, source lines 16227–16227. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $f : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ be a morphism of ringed spaces. Let $y \in Y$. Assume that*

(1) *f is closed,*
(2) *f is separated, and*
(3) *$f^{-1}(y)$ is quasi-compact.*

Then for E in $D^+(\mathcal{O}_X)$ we have $(Rf_*E)_y = R\Gamma(f^{-1}(y), E|_{f^{-1}(y)})$ in $D^+(\mathcal{O}_{Y,y})$.

Proof. The base change map of Lemma 02N7 gives a canonical map $(Rf_*E)_y \to R\Gamma(f^{-1}(y), E|_{f^{-1}(y)})$. To prove this map is an isomorphism, we represent E by a bounded below complex of injectives $\mathcal{I}^\bullet$. Set $Z = f^{-1}(\{y\})$. The assumptions of Lemma 09V3 are satisfied, see Topology, Lemma 0CY2. Hence the restrictions $\mathcal{I}^n|_Z$ are acyclic for $\Gamma(Z, -)$. Thus $R\Gamma(Z, E|_Z)$ is represented by the complex $\Gamma(Z, \mathcal{I}^\bullet|_Z)$, see Derived Categories, Lemma 015E. In other words, we have to show the map

$$\operatorname{colim}_V \mathcal{I}^\bullet(f^{-1}(V)) \longrightarrow \Gamma(Z, \mathcal{I}^\bullet|_Z)$$

is an isomorphism. Using Lemma 09V3 we see that it suffices to show that the collection of open neighbourhoods $f^{-1}(V)$ of $Z = f^{-1}(\{y\})$ is cofinal in the system of all open neighbourhoods. If $f^{-1}(\{y\}) \subset U$ is an open neighbourhood, then as f is closed the set $V = Y \setminus f(X \setminus U)$ is an open neighbourhood of y with $f^{-1}(V) \subset U$. This proves the lemma. $\square$